

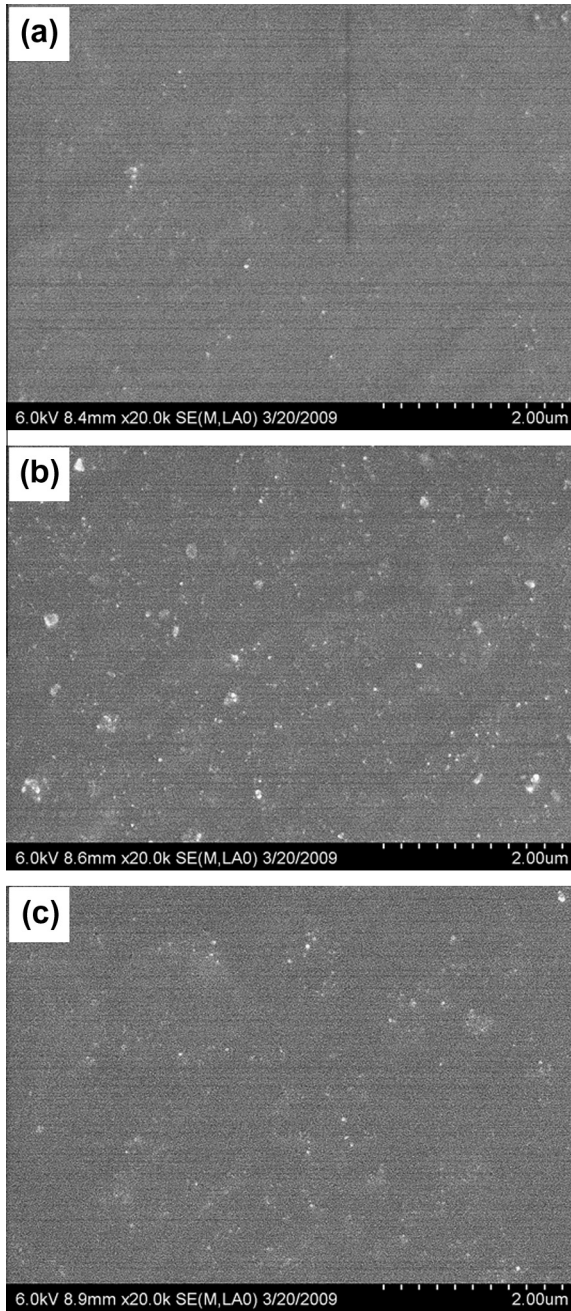


Fig. 6. SEM micrographs of fractured surface of PDMS networks filled with (a) 3 phr of R106, (b) 10 phr of R106 and (c) 10 phr of R8200.

chains adsorbed onto the silica surface increases. Moreover, the particle distance decreases with the increased number of fumed silica particles. Consequently, the number density of segments increases sharply and the length of segments decreases correspondingly, which can be verified by the reduction of upturn strain and equilibrium swelling ratio as listed in Table 2. These alternations of networks contribute to the increase of both effective volume effect and synergistic effect. It can be postulated that the percolation concentration for the formation of “hierarchical network” structure decreases with the specific surface area.

3.5. Effect of surface chemistry on tensile properties

Surface chemistry is of significant importance in the applications of fumed silica [53,54]. To investigate the effect of surface

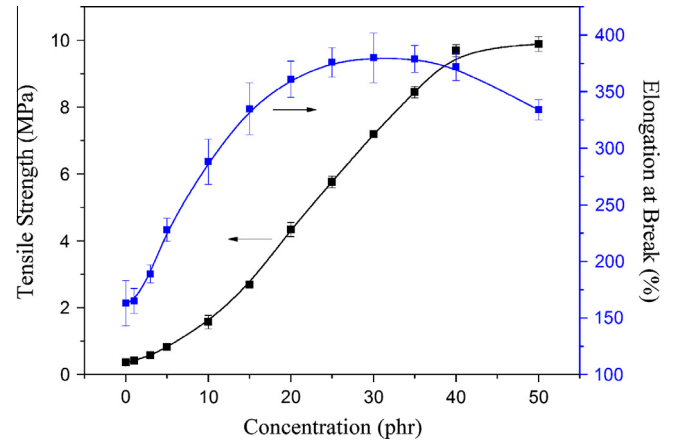


Fig. 7. Tensile properties for networks filled with R106 at a large scale of concentrations (Crosshead speed for testing: 500 mm/min). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

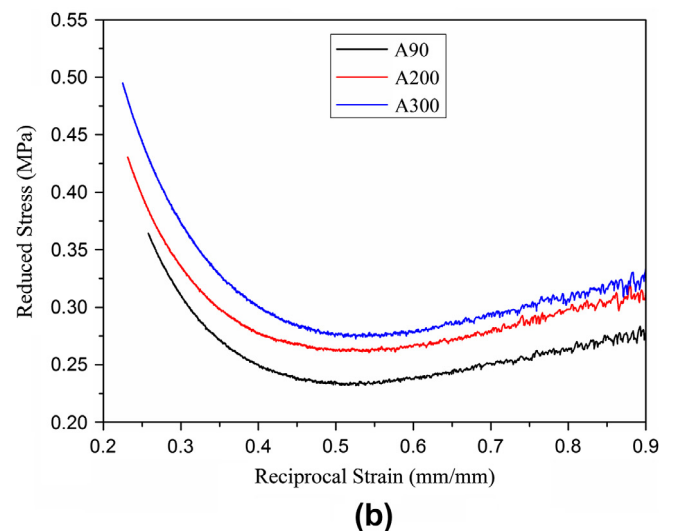
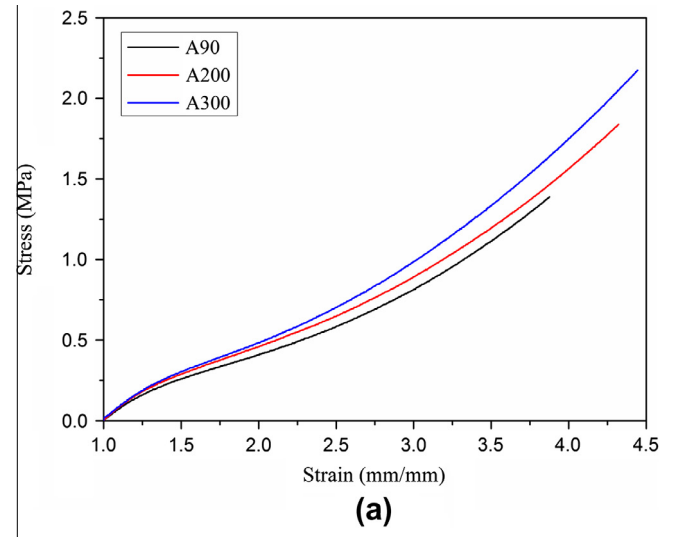


Fig. 8. Stress-strain curves (a) and Mooney-Rivlin plots and (b) for networks filled with fumed silica of different specific surface areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

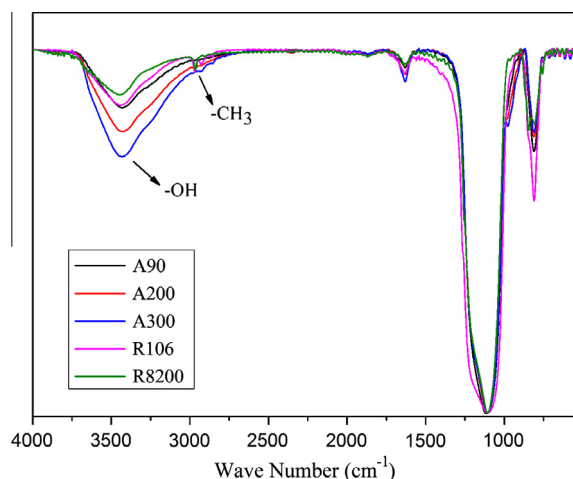


Fig. 9. Normalized IR spectra for different kinds of fumed silica. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

chemistry on tensile properties, three kinds of fumed silica were adopted: A300, R106 and R8200. Samples were labeled with the trade names of fumed silica.

From IR spectrum in Fig. 9, it is seen that after surface modification the peak standing for Si–OH group around 3500 cm^{-1} decreases to variant extents, meanwhile new peaks corresponding to $-\text{CH}_3$ groups between 2900 cm^{-1} and 3000 cm^{-1} emerge. It can also be inferred from the spectrum that R8200 is more fully modified compared with R106. Herein, we neglect the variance of particle number caused by modification and the influence of different modifiers, thus we can focus the discussion mainly on the extent of surface modification.

It is generally expected that networks filled with more fully modified fumed silica exhibit lower modulus and poorer ultimate tensile properties due to less effective volume effect and synergistic effect. This is the case for R8200 filled network as shown in Fig. 10. Although R8200 particles have a better dispersion state (refer to Fig. 6 (b and c)), the influence from the reduction of adsorption and entanglement is more predominant. The reduction can be verified by the increase of upturn strain and equilibrium swelling ratio as listed in Table 2.

However, for moderately modified R106 filled network, the ultimate tensile properties decrease slightly while the modulus increases compared with A300 filled network. This explanation is very complicated but can be better understood from the aspect of occluded rubber. Thanks to the excessive amount of silanol groups on the silica surface, the suspension filled 10 phr of A300 behaves more like a solid. After solution curing, a certain amount of occluded rubber forms within huge agglomerates. The occluded rubber can alter the network structure (refer to the upturn strain and equilibrium swelling ratio in Table 2) and thus affect both effective volume effect and synergistic effect. After surface modification, the dispersion of particles improves and the amount of occluded rubber decreases, which obviously has a positive influence on the tensile properties. It has to be mentioned here that the ultimate tensile properties can otherwise be improved after modification as was observed in our other experiments. The significance of moderate surface modification probably lies in reducing the viscosity of filled suspensions without dramatically damaging the mechanical properties of filled networks.

From this section we can figure out the significance of the extent of surface modification and the role of polymer–filler interaction played in reinforcement. It can also be inferred that the percolation concentration for the formation of “hierarchical network” structure increases after surface modification.

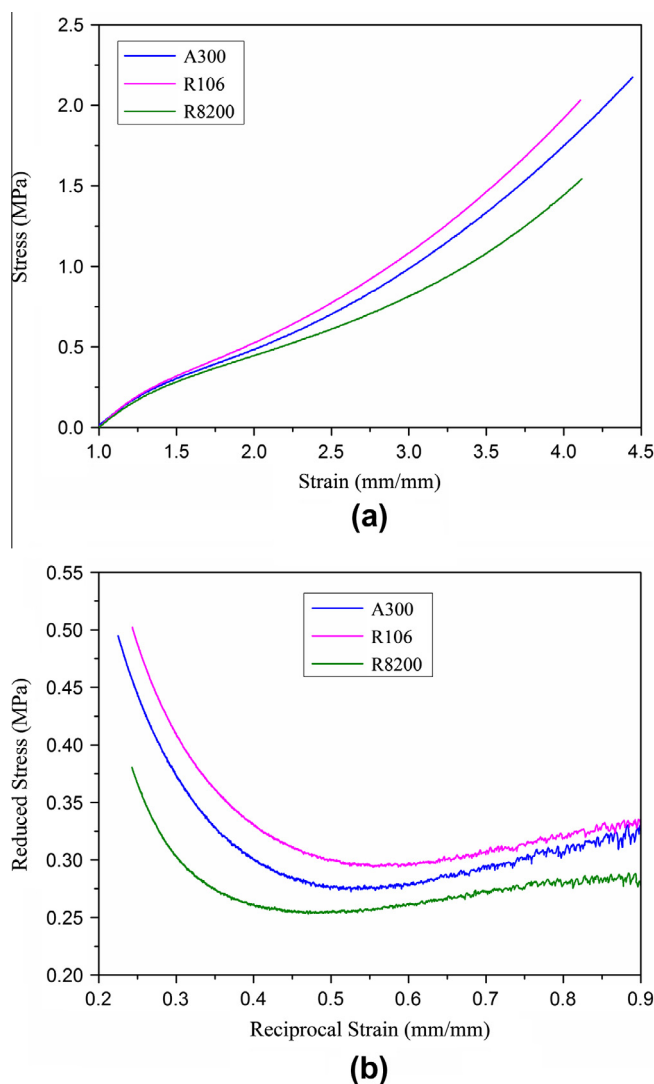


Fig. 10. Stress–strain curves (a) and Mooney–Rivlin plots (b) for networks filled with fumed silica of different surface chemistry. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Conclusions

We have investigated the tensile properties of fumed silica filled PDMS networks by analyzing the stress transfer of network chains under extension. Compared with unfilled networks, networks filled with fumed silica consist of network chains including segments and chains. The “hierarchical network” structure formed within highly filled networks (above the percolation concentration) sets a stage for the segments to exhibit non-Gaussian behavior before desorption occurs and for the networks to exhibit non-Gaussian effect. Reinforcing effect originates from the synergistic effect between network chains of different length scales in the “hierarchical network”, in addition to the effective volume effect originated from filler volume and polymer–filler interaction. The extent of synergistic effect, which can be characterized by the excess deformation energy of networks, is mainly determined by the number density and length distribution of network chains. It was showed that as dispersion improves, modulus decreases while ultimate tensile properties increase; as filler concentration and specific surface area increase, all mechanical parameters increase; tensile properties vary differently depending on the extent of surface modification. Experimental results can be explained by ana-

lyzing the variation of both effective volume effect and synergistic effect.

Acknowledgements

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